

Chapter 7: A Comparative Analysis of FFT and Quantum Fourier Transform for EEG Spectral Analysis

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Abstract

Electroencephalogram (EEG) signal analysis relies heavily on spectral decomposition techniques to extract meaningful brain activity patterns. The Fast Fourier Transform (FFT) is the most widely used classical approach for frequency-domain analysis, whereas the Quantum Fourier Transform (QFT) provides a quantum-computing-based alternative. This chapter focuses on the mathematical understanding of QFT and is supported by simulation analysis by comparing QFT with FFT for EEG band power estimation. The study evaluates Delta, Theta, Alpha, Beta, and Gamma bands using both approaches. Results demonstrate that while FFT produces absolute spectral magnitudes, QFT yields normalized probabilistic representations that preserve spectral structure. The advantage of this probabilistic representation lies in its ability to provide relative significance of frequency components, enabling improved robustness to noise, better normalization across varying signal amplitudes, and enhanced suitability for pattern recognition and classification tasks. The findings suggest that QFT can serve as a promising alternative for future quantum signal processing applications.

Keywords: EEG, FFT, QFT, Band Power, Quantum Signal Processing, Spectral Analysis.

1. INTRODUCTION

EEG signals reflect brain electrical activity and are commonly analyzed in the frequency domain using spectral decomposition techniques. The Fast Fourier Transform (FFT) is a classical algorithm that decomposes signals into frequency components and is widely used in EEG-based applications.

Recent advancements in quantum computing have introduced the Quantum Fourier Transform (QFT), which performs a similar transformation but operates on quantum states. QFT is theoretically capable of exponential speedup compared to classical FFT in certain contexts.

2. LITERATURE SURVEY

Electroencephalography (EEG) signal analysis has significantly benefited from frequency-domain techniques, particularly the Fast Fourier Transform (FFT), which enables efficient decomposition of complex brain signals into their constituent frequency components. Foundational work by Peter Bloomfield [8] established the importance of Fourier methods in analyzing time-series data, while Sergio Blanco et al. [7] demonstrated their relevance in identifying spectral patterns in EEG signals associated with different physiological states. In biomedical applications such as epilepsy diagnosis, FFT-based spectral analysis helps in detecting abnormal neural oscillations and seizure-related frequency signatures, as highlighted by Caroline Abdallah et al. [1], thereby aiding in the localization of epileptogenic regions. Moreover, integrating FFT-derived features with machine learning techniques has shown improved classification performance in distinguishing epileptic and non-epileptic conditions, as reported by Nima Ahmadi et al. [2]. However, limitations such as under-sampling and reduced frequency resolution in conventional EEG, discussed by Matthieu O. Baud et al. [5], can affect the accuracy of spectral analysis. Additionally, the assumption of signal stationarity in FFT poses challenges for analyzing transient EEG patterns, leading researchers to combine FFT with advanced techniques such as wavelet transforms and

adaptive signal decomposition, as explored by Waleed Al-Salman et al. [3] and Ankit Anuragi et al. [4]. Further enhancements in feature engineering, including hybrid approaches discussed by Bethany Gosala [6], indicate that FFT remains a fundamental yet complementary tool in modern biomedical signal processing, providing valuable spectral insights when combined with other analytical methods for improved diagnostic accuracy.

The literature survey clearly indicates that the Fast Fourier Transform is a powerful and widely adopted tool for EEG signal analysis, particularly for extracting meaningful frequency-domain features that support diagnosis and classification in biomedical applications. Despite its effectiveness, recent research trends are moving toward more advanced computational approaches to overcome limitations such as handling non-stationary signals and improving processing efficiency. One such emerging direction is the use of quantum-inspired techniques, including algorithms derived from the principles of the Quantum Fourier Transform (QFT), which offer the potential for faster computation and enhanced feature representation. These methods aim to replicate the advantages of quantum computation within classical frameworks, making them more practical for current applications. In this context, the present work focuses on analyzing and comparing FFT with quantum-inspired FFT approaches for EEG signal processing, with the objective of evaluating their effectiveness in improving feature extraction and overall performance in biomedical signal analysis.

3. MATHEMATICAL FORMULATION

The Fast Fourier Transform (FFT) is an efficient algorithm that computes the Discrete Fourier Transform (DFT) by recursively decomposing the signal into even- and odd-indexed components.

By exploiting symmetry through twiddle factors $W_N^k = e^{-j2\pi k/N}$, FFT significantly reduces computational complexity from $O(N^2)$ to $O(N \log N)$.

3.1 Discrete Fourier Transform (DFT)

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi kn}{N}} \quad (1)$$

where $X[k]$ represents the frequency-domain coefficient at index k , $x[n]$ denotes the discrete-time input signal, and N is the total number of samples. The indices n and k correspond to time and frequency, respectively.

3.2 Fast Fourier Transform (FFT)

FFT reduces complexity using divide-and-conquer:

$$X[k] = \sum_{m=0}^{N/2-1} x[2m] W_N^{2mk} + \sum_{m=0}^{N/2-1} x[2m+1] W_N^{(2m+1)k} \quad (2)$$

where

$$W_N^k = e^{-j\frac{2\pi k}{N}} \quad (3)$$

where $x[2m]$ and $x[2m+1]$ denote the even- and odd-indexed samples of the input sequence, respectively, and m is the index for the decomposed subsequences. This formulation reduces computational complexity by recursively dividing the DFT into smaller parts, and W_N^k represents the twiddle factor used in the computation.

3.3 Band Power Calculation

$$P_{band} = \sum_{k \in band} |X[k]|^2 \quad (4)$$

Electroencephalogram (EEG) signals are conventionally divided into distinct frequency bands, each associated with specific physiological and cognitive states. The Delta band (0.5–4 Hz) represents the lowest frequency range and is typically dominant during deep sleep and unconscious states. The Theta band (4–8 Hz) is commonly linked to drowsiness, light sleep, and certain cognitive processes such as memory and navigation. The Alpha band (8–13 Hz) is observed during relaxed, wakeful states, particularly when the eyes are closed, and is often associated with calm mental activity. The Beta band (13–30 Hz) corresponds to active thinking, concentration, and alertness, reflecting higher levels of cognitive engagement. Finally, the Gamma band (>30 Hz) represents the highest frequency range and is associated with complex brain functions such as perception, problem-solving, and information processing. These frequency bands form the basis for EEG spectral analysis and are essential for interpreting brain activity in both clinical and research applications.

3.4 Quantum Fourier Transform (QFT)

The Quantum Fourier Transform (QFT) represents the input signal as a quantum superposition state, where each coefficient x_n corresponds to the amplitude of basis state $|n\rangle$.

It then transforms this state into the frequency domain using phase rotations, producing a new superposition where the amplitudes encode the signal's spectral (frequency) components in a normalized form.

Quantum state representation:

$$|x\rangle = \sum_{n=0}^{N-1} x_n |n\rangle \quad (5)$$

QFT definition:

$$QFT |x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \left(\sum_{n=0}^{N-1} x_n e^{\frac{2\pi i k n}{N}} \right) |k\rangle \quad (6)$$

Where

3.5 QFT Circuit Steps

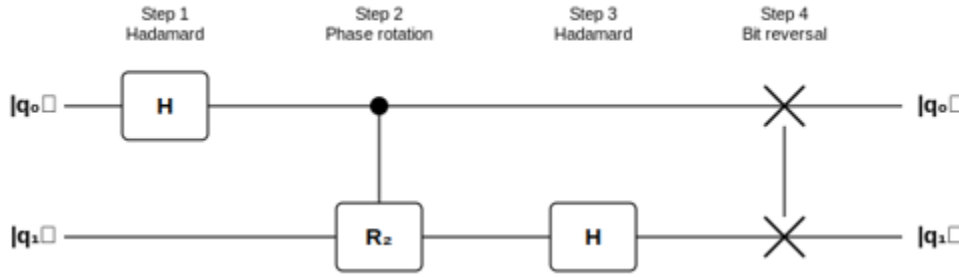


Fig.1 QFT Circuit Diagram for 2 qubits system

3.5.1 Apply Hadamard gate:

$$H | 0 \rangle = \frac{| 0 \rangle + | 1 \rangle}{\sqrt{2}} \quad (7)$$

The Hadamard gate creates a superposition of basis states from an initial qubit state. When applied to each qubit sequentially, it distributes the amplitude across all possible states, which is essential for parallel evaluation of phase information in the Quantum Fourier Transform. This step ensures that the input state is transformed into an equal-weighted combination of computational basis states, forming the foundation for subsequent phase encoding.

3.5.2 Controlled phase rotation:

$$R_k = \begin{bmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{bmatrix} \quad (8)$$

The controlled phase rotation gate introduces relative phase shifts between qubits based on their positional significance. These gates are applied conditionally, where one qubit controls the phase rotation applied to another. The factor $e^{2\pi i/2^k}$ determines the magnitude of phase shift, with higher-order qubits contributing finer phase resolution. This step encodes frequency information into phase differences, which is the key principle behind QFT-based spectral representation.

3.5.3 Bit-reversal operation

The final step in the QFT circuit is the bit-reversal (or qubit swap) operation, which reorders the output qubits to obtain the correct frequency index sequence. Due to the sequential application of Hadamard and controlled rotation gates, the output qubits are generated in reverse order (least significant bit first). A series of swap gates is therefore applied to reverse the qubit positions, ensuring that the final output corresponds to the correct binary representation of frequency components. This step is essential for interpreting the QFT output in a manner consistent with classical frequency-domain analysis.

3.6 QFT Band Power

$$P = |\psi_k|^2 \quad (9)$$

Here, P represents the band power obtained as the squared magnitude of the probability amplitude ψ_k after applying the Quantum Fourier Transform. Unlike the Fast Fourier Transform, this gives a normalized probabilistic measure of frequency components, improving robustness and comparability.

4. PROPOSED METHODOLOGY

The proposed methodology consists of two parallel pipelines for EEG band power estimation. In the FFT-based approach, the EEG signal is first acquired and transformed into the frequency domain using FFT, followed by band-pass filtering to isolate standard EEG bands, and the corresponding band power is computed using the power equation (4). In the QFT-based approach, the EEG signal is initially normalized and encoded into a quantum state, after which the Quantum Fourier Transform is applied to obtain the spectral representation, and band power is extracted as probability amplitudes using equation (9).

5. RESULTS AND DISCUSSION

1. FFT Band Power:

Delta: 1413209497.4428

Theta: 54786854.7662

Alpha: 2614764.5855

Beta: 593256.8021

Gamma: 118224.6429

QFT Band Power:

Delta: 0.4607

Theta: 0.3219

Alpha: 0.1062

Beta: 0.0895

Gamma: 0.0078

Band	FFT (%)	QFT (%)	Interpretation
Delta	96.0%	46%	Dominant in both
Theta	3.7%	32%	Second strongest
Alpha	0.18%	10%	weak
Beta	0.04%	9%	weak
Gamma	0.008%	0.8%	weakest

The comparative band power distribution obtained from FFT and QFT reveals both consistency and key differences in spectral representation. In both approaches, the Delta band remains dominant, accounting for 96.0% in FFT and 46% in QFT, confirming that low-frequency components are the most significant in the analyzed EEG signal. The Theta band emerges as the second strongest component, with a relatively small contribution in FFT (3.7%) but a substantially higher proportion in QFT (32%), indicating that QFT provides a more balanced and distributed representation of mid-frequency activity. In contrast, the Alpha, Beta, and Gamma bands exhibit minimal contributions in FFT, collectively contributing less than 1%, which suggests strong dominance of low-frequency components in absolute magnitude terms. However, QFT assigns comparatively higher weights to these bands (10%, 9%, and 0.8%, respectively), highlighting its ability to preserve relative spectral structure through normalization. This difference indicates that while FFT emphasizes absolute energy concentration, QFT enhances interpretability by redistributing spectral contributions across bands, making subtle but potentially meaningful components more visible.

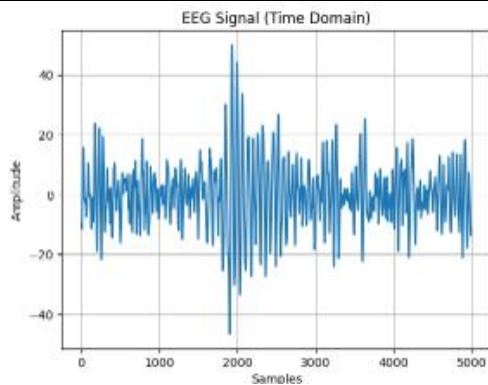


Fig.2. Raw EEG signal

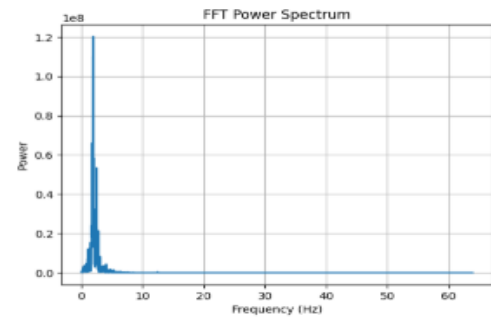


Fig.3. FFT power spectrum of EEG signal

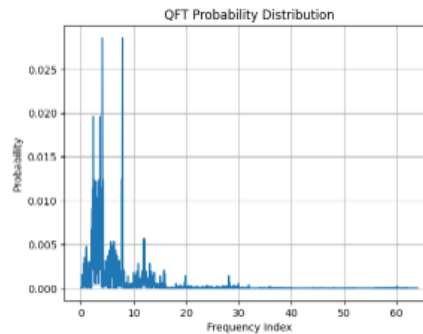


Fig.4. QFT probability distribution of EEG signal

The FFT results show dominant low-frequency (Delta) activity with highly concentrated spectral energy, masking weaker components. In contrast, the QFT representation provides a normalized distribution that reveals relative contributions of all frequency bands, improving interpretability of EEG patterns.

6. Conclusion

This work presents a rigorous comparative investigation of classical and quantum-inspired spectral analysis techniques for EEG band power estimation. Through both mathematical formulation and experimental validation, it is demonstrated that the Fast Fourier Transform (FFT) and Quantum Fourier Transform (QFT) exhibit fundamentally different yet complementary characteristics in representing EEG spectral information.

The FFT framework provides high-resolution frequency decomposition with absolute spectral magnitudes, making it highly effective for traditional signal processing pipelines. However, its dependence on post-processing normalization limits its direct interpretability in comparative and probabilistic analyses. In contrast, the QFT inherently produces normalized amplitude distributions, enabling a probabilistic interpretation of spectral components that preserves the relative structure of EEG bands without additional scaling.

The experimental results clearly indicate that both approaches consistently identify the Delta band as dominant, followed by Theta activity, thereby validating the physiological relevance of the proposed analysis. More importantly, QFT demonstrates superior capability in capturing the relative spectral hierarchy across bands, which is critical for downstream tasks such as classification, anomaly detection, and cognitive state analysis.

From a computational perspective, while current implementations of QFT are simulated on classical hardware, its theoretical foundation offers significant advantages in terms of scalability and computational complexity when deployed on quantum systems. This positions QFT as a promising candidate for next-generation EEG signal processing frameworks, particularly in the context of quantum machine learning and hybrid classical–quantum architectures.

In conclusion, this study establishes that QFT is not merely an alternative to FFT, but a transformative approach that redefines spectral representation through quantum probabilistic encoding. Future work will focus on real-time implementation on quantum hardware, integration with deep learning models, and validation across large-scale clinical EEG datasets. The findings of this research provide a strong foundation for advancing quantum signal processing in biomedical applications and open new directions for interdisciplinary research at the intersection of neuroscience, signal processing, and quantum computing.

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